

Heart Failure — HFrEF drugs (TABLE 265-2)

TABLE 265-2 Guideline-Directed Pharmacologic Therapy and Target Doses in Heart Failure with Reduced Ejection Fraction				
DRUG CLASS	GENERIC DRUG	MEAN DAILY DOSE IN CLINICAL TRIALS (mg)	INITIATION (mg)	TARGET DOSE (mg)
Angiotensin-Converting Enzyme Inhibitors				
	Lisinopril	4.5–33	2.5–5 qd	20–35 qd
	Enalapril	17	2.5 bid	10–20 bid
	Captopril	123	6.25 tid	50 tid
	Trandolapril	N/A	0.5–1 qd	4 qd
Angiotensin Receptor Blockers				
	Losartan	129	50 qd	150 qd
	Valsartan	254	40 bid	160 bid
	Candesartan	24	4–8 qd	32 qd
Aldosterone Antagonists				
	Eplerenone	42.6	25 qd	50 qd
	Spironolactone	26	12.5–25 qd	25–50 qd
Beta Blockers				
	Metoprolol succinate CR/XL	159	12.5–25 qd	200 qd
	Carvedilol	37	3.125 bid	25–50 bid
	Bisoprolol	8.6	1.25 qd	10 qd
Arteriovenous Vasodilators				
	Hydralazine isosorbide dinitrate	270/136	37.5/20 tid	75/40 tid
	Fixed-dose hydralazine/isosorbide dinitrate	143/76	37.5/20 qid	75/40 qid
Angiotensin Receptor-Neprilysin Inhibitor				
	Sacubitril-valsartan	375	100 bid	200 bid
SGLT-2 Inhibitor				
	Dapagliflozin	10	10 qd	10 qd
	Empagliflozin	10	10 qd	10 qd
	Sotagliflozin	200	200 qd	200 qd
Novel Therapies				
	Vericiguat (sGC stimulator)	9.2	2.5 qd	10 qd
	Omecamtiv mecarbil (myosin activator)	Not reported	25 bid	Up to 50 mg bid (based on plasma concentrations)

1. ACE inhibitors

WHAT YOU SEE

Row 'Angiotensin-Converting Enzyme Inhibitors'

WHAT IT ENCODES

ACE inhibitors (lisinopril, enalapril, captopril, trandolapril) block conversion of angiotensin I→II, reducing afterload, aldosterone and adverse remodeling — a foundational mortality-reducing RAAS therapy in HFrEF (SOLVD, CONSENSUS). Start low, up-titrate to target, monitor K+ and creatinine, and accept a modest (~<30%) creatinine rise.

BOARD TRAP

WRONG: continue the ACEi and add an ARB or ARNI on top for 'extra RAAS blockade.' Discriminator: never combine ACEi with ARB or ARNI (hyperkalemia, hypotension, renal failure, angioedema) — and require a ≥36-hour washout when switching ACEi→ARNI to avoid angioedema.

2. Lisinopril target 20-35mg

WHAT YOU SEE

Lisinopril row, target 20-35 qd

WHAT IT ENCODES

Up-titrate lisinopril (initiation 2.5–5 mg) toward its target 20–35 mg once daily, because the mortality/morbidity benefit of RAAS inhibition is dose-dependent (ATLAS: high-dose lisinopril reduced hospitalizations vs low dose). Don't leave patients stranded at the starting dose.

BOARD TRAP

WRONG: any ACEi dose that lowers blood pressure is good enough. Discriminator: ATLAS showed the higher target dose, not just 'on an ACEi,' drives the outcome benefit — under-titration is a common quality gap, so push to target as BP/renal function allow.

3. ARBs

WHAT YOU SEE

Row 'Angiotensin Receptor Blockers'

WHAT IT ENCODES

ARBs (losartan target 150 mg, valsartan 160 mg bid, candesartan 32 mg) block the AT1 receptor and are used when an ACEi is not tolerated, classically due to ACEi-induced cough (bradykinin) — CHARM-Alternative, Val-HeFT. They are second-line to ARNI/ACEi for RAAS inhibition in HFrEF.

BOARD TRAP

WRONG: a patient who had ANGIOEDEMA on an ACEi can safely switch to an ARB. Discriminator: ARBs carry a small cross-reactive angioedema risk and are used cautiously/avoided after ACEi angioedema — cough (not angioedema) is the clean reason to switch ACEi→ARB.

4. Aldosterone antagonists (MRA)

WHAT YOU SEE

Row 'Aldosterone Antagonists'

WHAT IT ENCODES

Mineralocorticoid receptor antagonists (eplerenone, spironolactone) block aldosterone, reducing fibrosis, sodium retention and mortality in HFrEF (RALES spironolactone, EMPHASIS-HF/EPHESUS eplerenone). Indicated for EF ≤35% on ACEi/ARB + beta-blocker. Require K+ <5.0 and eGFR >30 to start; recheck K+/creatinine within 1 week.

BOARD TRAP

WRONG: start an MRA in a HFrEF patient with eGFR 25 or K+ 5.4. Discriminator: MRAs are contraindicated when eGFR <30 or K+ >5.0 — and combining an MRA with an ACEi/ARNI plus a potassium supplement or trimethoprim is a classic life-threatening hyperkalemia setup.

5. Spironolactone vs eplerenone

WHAT YOU SEE

Spironolactone target 25-50, eplerenone target 50

WHAT IT ENCODES

Spironolactone (target 25–50 mg qd) is non-selective and binds androgen/progesterone receptors, causing gynecomastia, breast tenderness and menstrual changes. Eplerenone (target 50 mg qd) is selective for the mineralocorticoid receptor, so it largely avoids those endocrine side effects — switch to eplerenone if spironolactone causes painful gynecomastia.

BOARD TRAP

WRONG: gynecomastia from spironolactone means you must stop all MRA therapy. Discriminator: switch to selective eplerenone rather than abandoning the class — MRAs are mortality-reducing, and the endocrine effects are a spironolactone-specific, fixable problem.

6. Beta blockers

WHAT YOU SEE

Row 'Beta Blockers'

WHAT IT ENCODES

Beta-blockers counter chronic sympathetic toxicity in HFrEF, reducing mortality and producing reverse remodeling — but ONLY three are proven: carvedilol, metoprolol SUCCINATE (CR/XL), and bisoprolol (CIBIS-II, MERIT-HF, COPERNICUS). Start low, go slow, and only when the patient is euvolemic/compensated (they can transiently worsen HF).

BOARD TRAP

WRONG: all beta-blockers in this class are interchangeable for HFrEF. Discriminator: only carvedilol, metoprolol succinate, and bisoprolol have mortality evidence — metoprolol TARTRATE, atenolol and others do NOT count, and substituting tartrate for succinate is a frequently tested error.

7. Metoprolol succinate CR/XL target 200

WHAT YOU SEE

Metoprolol succinate CR/XL, target 200 qd

WHAT IT ENCODES

Use metoprolol SUCCINATE CR/XL (extended-release, once daily), NOT metoprolol tartrate (immediate-release, twice daily). Initiation 12.5–25 mg, target 200 mg daily (MERIT-HF). The succinate formulation gives the steady beta-1 blockade that delivered the mortality benefit.

BOARD TRAP

WRONG: metoprolol tartrate twice daily is equivalent to metoprolol succinate for HFrEF. Discriminator: only the SUCCINATE (CR/XL) salt was studied and proven in MERIT-HF — tartrate has no HFrEF mortality evidence; the salt/formulation is the whole point of this row.

8. Carvedilol / bisoprolol dosing

WHAT YOU SEE

Carvedilol 25-50 bid, bisoprolol 10 qd

WHAT IT ENCODES

Carvedilol (initiation 3.125 mg bid, target 25 mg bid; 50 mg bid if >85 kg) is a non-selective beta- plus alpha-blocker (extra vasodilation, helpful in HTN). Bisoprolol (initiation 1.25 mg, target 10 mg qd) is beta-1 selective (CIBIS-II) — a good choice when alpha-blockade/hypotension or reactive airways are a concern.

BOARD TRAP

WRONG: avoid all beta-blockers in a HFrEF patient with mild COPD/asthma. Discriminator: a cardioselective agent (bisoprolol or metoprolol succinate) is generally tolerated and the mortality benefit outweighs the small bronchospasm risk — carvedilol (non-selective) is the one to avoid in significant reactive airway disease.

9. Hydralazine + ISDN

WHAT YOU SEE

Row 'Arteriovenous Vasodilators' hydralazine isosorbide dinitrate

WHAT IT ENCODES

Hydralazine (arterial/afterload reducer) + isosorbide dinitrate (venodilator/preload, nitric-oxide donor) — target ~75/40 mg tid or the fixed-dose combo. Add to GDMT in self-identified BLACK patients with NYHA III–IV HFrEF (A-HeFT), and use as the RAAS alternative when ACEi/ARB/ARNI are contraindicated (renal failure, hyperkalemia, hypotension).

BOARD TRAP

WRONG: hydralazine-ISDN can replace ACEi/ARB/ARNI as routine first-line RAAS therapy in everyone. Discriminator: it is ADD-ON therapy in Black patients on top of standard GDMT (A-HeFT), or a substitute only when RAAS inhibitors can't be used — it does not block the RAAS and is not a universal first-line agent.

10. ARNI (sacubitril-valsartan)

WHAT YOU SEE

Row 'Angiotensin Receptor-Neprilysin Inhibitor' sacubitril-valsartan

WHAT IT ENCODES

ARNI = sacubitril (neprilysin inhibitor → raises beneficial natriuretic peptides) + valsartan (ARB). Initiation 24/26–49/51 mg bid, target 97/103 mg (~200 mg) bid. PARADIGM-HF showed ARNI SUPERIOR to enalapril for CV death/HF hospitalization, so it is the PREFERRED RAAS agent in HFrEF.

BOARD TRAP

WRONG: start ARNI the day after stopping an ACEi (or while still on it). Discriminator: you MUST allow a ≥36-hour washout from an ACEi before starting ARNI because overlapping neprilysin + ACE inhibition causes angioedema — also avoid ARNI with eGFR/hyperkalemia issues, and it raises BNP but lowers NT-proBNP.

11. SGLT2 inhibitor (fixed 10mg)

WHAT YOU SEE

Row 'SGLT-2 Inhibitor' dapagliflozin/empagliflozin 10mg

WHAT IT ENCODES

SGLT2 inhibitors (dapagliflozin, empagliflozin) reduce CV death and HF hospitalization in HFrEF regardless of diabetes (DAPA-HF, EMPEROR-Reduced) via natriuresis and cardiac-metabolic effects. Fixed 10 mg once daily — NO titration. Sotagliflozin (SGLT1/2, target 200 mg) is an alternative. Watch for euglycemic DKA, volume depletion, and genital mycotic infections.

BOARD TRAP

WRONG: SGLT2i must be up-titrated to a target dose like the other pillars. Discriminator: it is a FIXED 10 mg dose with no titration — and unlike its diabetes indication, it is given for HFrEF outcomes even in non-diabetics; do not withhold it for a normal HbA1c.

12. Novel: vericiguat

WHAT YOU SEE

'Vericiguat (sGC stimulator) target 10mg qd'

WHAT IT ENCODES

Vericiguat is a soluble guanylate cyclase (sGC) STIMULATOR that augments the NO–cGMP pathway when endogenous NO is low. Initiation 2.5 mg, target 10 mg qd. VICTORIA: in high-risk HFrEF with a recent worsening event, it modestly reduced CV death/HF hospitalization — an add-on, not a core pillar.

BOARD TRAP

WRONG: vericiguat is a first-line pillar that replaces the SGLT2i or MRA. Discriminator: it is reserved as ADD-ON for high-risk patients with a recent decompensation already on GDMT (VICTORIA) — and don't combine with nitrates/PDE5i (additive hypotension via the same cGMP pathway).

13. Novel: omecamtiv mecarbil

WHAT YOU SEE

'Omecamtiv mecarbil (myosin activator)'

WHAT IT ENCODES

Omecamtiv mecarbil is a selective cardiac MYOSIN activator that increases contractility (systolic ejection time) without raising myocardial oxygen demand or intracellular calcium. Dosed by plasma level (up to ~50 mg bid). GALACTIC-HF: small reduction in HF events but no mortality benefit — investigational/limited role.

BOARD TRAP

WRONG: omecamtiv mecarbil works like an inotrope such as milrinone or dobutamine and raises myocardial oxygen demand. Discriminator: it improves contractility WITHOUT increasing oxygen consumption or arrhythmia the way calcium-dependent inotropes do — but its clinical benefit is modest and it is not standard GDMT.

14. Initiation vs target columns

WHAT YOU SEE

Headers 'INITIATION' and 'TARGET DOSE'

WHAT IT ENCODES

The table is built around two columns: the low INITIATION dose (where you start to avoid hypotension/renal hits/hyperkalemia) and the evidence-based TARGET DOSE (where the trial mortality benefit was achieved). The principle: start low, titrate up to target as BP, K+ and renal function allow.

BOARD TRAP

WRONG: keeping a patient 'on all four pillars' at starting doses is optimized GDMT. Discriminator: outcomes track titration to target doses — but the modern STRONG-HF approach gets all four classes ON first at low doses, then rapidly up-titrates, rather than maxing one drug before adding the next.

15. Avoid in HFrEF — non-DHP CCB, NSAIDs, TZDs

WHAT YOU SEE

Drugs-to-avoid box: verapamil/diltiazem, NSAIDs, glitazones

WHAT IT ENCODES

Drugs that WORSEN HFrEF and should be avoided: non-dihydropyridine CCBs (verapamil, diltiazem) — negative inotropes that increase mortality; NSAIDs — cause sodium/water retention, blunt diuretics/RAAS and precipitate decompensation/AKI; thiazolidinediones (pioglitazone, rosiglitazone) — fluid retention worsening HF; and class I antiarrhythmics/dronedarone (pro-arrhythmic in low EF).

BOARD TRAP

WRONG: use diltiazem or verapamil for rate control of AF in a HFrEF patient. Discriminator: non-DHP CCBs are negative inotropes contraindicated in HFrEF — use a beta-blocker (or digoxin) for rate control instead; amlodipine/felodipine are the only CCBs considered neutral/safe in HFrEF.

16. Loop diuretics — symptom control

WHAT YOU SEE

Diuretic bottle: furosemide / torsemide / bumetanide

WHAT IT ENCODES

Loop diuretics (furosemide, torsemide, bumetanide) inhibit the Na-K-2Cl transporter to relieve congestion — they improve SYMPTOMS but have no proven mortality benefit. Torsemide/bumetanide have more reliable oral absorption than furosemide. IV-to-PO furosemide is roughly 2:1; equivalence $\approx$ furosemide 40 = torsemide 20 = bumetanide 1 mg.

BOARD TRAP

WRONG: loop diuretics reduce mortality and are a 'pillar' of HFrEF therapy. Discriminator: they are for congestion/symptom control only, not survival — the four pillars (ARNI, beta-blocker, MRA, SGLT2i) reduce mortality, and diuretics should be titrated to the lowest dose maintaining euvolemia.

17. Ivabradine & digoxin — adjuncts

WHAT YOU SEE

Adjunct row: ivabradine (SHIFT) and digoxin (DIG)

WHAT IT ENCODES

Ivabradine blocks the SA-node funny (If) current to slow rate WITHOUT affecting contractility — add when in SINUS rhythm with HR $\geq$ 70 despite a maximally tolerated beta-blocker, EF $\leq$ 35% (SHIFT: fewer HF hospitalizations). Digoxin (target trough 0.5–0.9 ng/mL) reduces HF hospitalizations but not mortality (DIG trial) — a late add-on for symptoms or AF rate control.

BOARD TRAP

WRONG: ivabradine works in atrial fibrillation and substitutes for a beta-blocker. Discriminator: ivabradine requires SINUS rhythm and is added ON TOP of a beta-blocker (not instead of it). For digoxin, watch toxicity (nausea, visual halos, arrhythmia) — risk rises with hypokalemia and renal failure, and the target trough is LOW (0.5–0.9).